Operacion Nexo 5G/6G - Informe tecnico

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Resumen

Este informe desarrolla el reto Operacion Nexo 5G/6G para Nueva Pangea comparando dos escenarios de alta demanda: un distrito financiero urbano y un festival masivo temporal.

El escenario A queda limitado por capacidad y recomienda N=4; el escenario B queda claramente limitado por densidad y exige cell splitting hasta la etapa S4.

1. Introduccion

El problema de una red movil moderna no se resuelve con un unico numero de cobertura. Una ciudad inteligente obliga a distinguir entre alcance radio, carga de trafico e interferencia. El objetivo del reto es demostrar con base fisica y matematica que la solucion de red debe cambiar cuando cambia la casuistica operativa.

2. Estado del arte

La evolucion hacia LTE y 5G consolida OFDMA, programacion dinamica de recursos y adaptacion MCS. En paralelo, los modelos de propagacion urbanos y suburbanos siguen siendo la base del dimensionamiento inicial. Para el plano de capacidad, Erlang B sigue siendo una herramienta docente valida cuando se desea estimar bloqueo y area maxima por celda.

2.1 Contexto radio moderno

modulation	design_tradeoff	typical_use	interpretation
QPSK	High robustness	Cell edge and difficult propagation	Lower spectral efficiency, lower SNR need
16QAM	Balanced	Typical urban load	Intermediate throughput and robustness
64QAM	High throughput	Good SINR zones	Higher spectral efficiency at higher SNR
256QAM	Peak efficiency	Excellent channel quality	Representative of modern LTE/5G adaptation logic

3. Metodologia

La metodologia reproduce exactamente la secuencia pedida en la guia docente: ruido, sensibilidad, perdida maxima, radio por cobertura, trafico por usuario, densidad de trafico, canales utiles, capacidad Erlang B y radio por capacidad. El criterio final adopta siempre el menor radio entre cobertura y capacidad.

3.1 Parametros base del sistema

parameter	value	engineering_role
Operating frequency	1800 MHz	Shared by both scenarios
System bandwidth	20 MHz	Used for thermal noise

Base station transmit power	43 dBm	Equivalent to 20 W
TX antenna gain	18 dBi	Base station sector antenna
RX antenna gain	0 dBi	Mobile terminal
Additional losses	12 dB	Cables, connectors and design margin
Receiver noise figure	7 dB	NF for sensitivity estimate
Implementation losses	2 dB	L_impl term in sensitivity
Blocking probability	2.00%	GoS objective for Erlang B
Total physical channels	100	Simplified PRB-like capacity pool

3.2 Objetivos por escenario

scenario	environment	density	design_priority	required_analysis
A_distrito_financiero	Dense urban core	2500 active users/km2	Coverage plus strong capacity pressure	Three sectors per site and N = 3, 4, 7 comparison
B_festival_global	Open suburban esplanade	8000 active users/km2	Extreme temporal congestion	Omnidirectional cells plus cell splitting evaluation

4. Escenarios

4.1 Escenario A - Distrito financiero

El distrito financiero parte de 250.0 Erl/km2, una SNR requerida de 15 dB y una arquitectura de tres sectores por sitio con analisis de N = 3, 4 y 7.

4.2 Escenario B - Festival global

El festival parte de 666.7 Erl/km2, una SNR requerida de 5 dB, celdas omnidireccionales y evaluacion explicita de cell splitting.

5. Resultados

5.1 Cobertura

scenario	thermal_noise_dbm	receiver_sensitivity_dbm	max_path_loss_db	coverage_radius_km	traffic_per_user_erlang	traffic_density_erlang_km2	target_area_km2	propagation_model
A_distrito_financiero	-93.9897	-76.9897	125.99	0.552793	0.1	250	1	L_p = 135 + 35 log10(d)
B_festival_global	-93.9897	-86.9897	135.99	3.41185	0.0833333	666.667	1	L_p = 120 + 30 log10(d)

5.2 Capacidad en el distrito financiero

reuse_factor_n	reuse_ratio_d_over_r	reuse_distance_km_if_design_radius	channels_per_site	channels_per_sector	sector_capacity_erlang	site_capacity_erlang	capacity_area_km2	capacity_radius_km	coverage_radius_km	design_radius_km	design_area_km2	sectorized_sir_db	omnidirectional_sir_db

sir_margin_db | limiting_factor | sites_for_target_area | recommended |
| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 3 | 3 | 0.492776 | 33 | 11 | 5.84153 | 17.5246 | 0.0700984 | 0.164259 | 0.552793 | 0.164259 | 0.0700984 | 13.6889 | 8.91773 | -1.31106 | capacity | 15 | False |
| 4 | 3.4641 | 0.448365 | 25 | 8 | 3.62705 | 10.8812 | 0.0435246 | 0.129432 | 0.552793 | 0.129432 | 0.0435246 | 15.8754 | 11.1042 | 0.875372 | capacity | 23 | True |
| 7 | 4.58258 | 0.32549 | 14 | 4 | 1.09226 | 3.27678 | 0.0131071 | 0.0710277 | 0.552793 | 0.0710277 | 0.0131071 | 20.1285 | 15.3573 | 5.12854 | capacity | 77 | False |

5.3 Capacidad y splitting en el festival

channels_per_cell	cell_capacity_erlang	capacity_area_km2	capacity_radius_km	coverage_radius_km	design_radius_km	design_area_km2	limiting_factor	sites_for_target_area
100	87.972	0.131958	0.225368	3.41185	0.225368	0.131958	capacity	8

split_stage	radius_km	area_km2	cells_per_original_footprint	capacity_density_erlang_km2	supported_users_km2	demand_users_km2	meets_demand	sites_for_target_area	recommended
0	3.41185	30.2435	1	2.90879	34.9055	8000	False	1	False
1	1.70593	7.56087	4	11.6352	139.622	8000	False	1	False
2	0.852963	1.89022	16	46.5407	558.488	8000	False	1	False
3	0.426481	0.472555	64	186.163	2233.95	8000	False	3	False
4	0.213241	0.118139	256	744.65	8935.81	8000	True	9	True
5	0.10662	0.0295347	1024	2978.6	35743.2	8000	True	34	False
6	0.0533102	0.00738366	4096	11914.4	142973	8000	True	136	False

6. Discusion

En el distrito financiero, el radio por cobertura alcanza 0.553 km, pero el radio por capacidad recomendado cae a 0.129 km. Esto demuestra que la celda se dimensiona por carga y no por alcance.

En el festival, la diferencia es todavia mas extrema: la cobertura teorica es 3.412 km mientras que la capacidad util solo sostiene 0.225 km. El problema dominante es la densidad de usuarios.

7. Supuestos y trazabilidad

assumption	applied_value	justification
Reference target area	1 km2 for both scenarios	The statement asks for the number of cells over a proposed target area but does not provide a fixed value. A common normalized 1 km2 area is assumed to compare cell density without bias.
Channel partition in scenario A	floor(100 / N) channels per site, then uniform split across 3 sectors	The statement requests N = 3, 4, 7 and three sectors per site, but it does not define a more detailed PRB scheduler. The uniform split is the simplest reproducible assumption.
Interference approximation	2 first-tier interferers for sectorized cells and 6 for omnidirectional reference	This is the standard first-tier hexagonal approximation used to compare reuse strategies in teaching-oriented planning exercises.

Hexagonal geometry	Regular hexagonal cell area model	Needed to convert traffic-limited area into an equivalent cell radius and then into the number of required sites.
Modern radio context	OFDMA/LTE/5G design logic with adaptive MCS	The statement explicitly asks for a modern mobile-network framing even though the numerical exercise is simplified to link budget and Erlang B.

8. Validacion frente a la rubrica

rubric_area	project_evidence	why_it_matters
Structure and presentation	Static web includes summary, scenarios, assumptions and code section	Supports rubric presentation block
State of the art	Project exports a modulation context table and OFDMA framing	Connects calculations with modern mobile systems
Methodology	Main pipeline reproduces noise, sensitivity, coverage, capacity, reuse and splitting	Step-by-step reproducibility is explicit
Mathematical precision	Outputs keep units, formulas and CSV traces	Facilitates annex verification
Discussion	Root web explains limiting factors and design trade-offs	Moves beyond raw numbers
Conclusions and future view	Report builder includes recommendations and 2030 extension note	Aligns with final rubric block

9. Conclusiones

La solucion adoptada para el distrito financiero es N=4 con tres sectores por sitio. La solucion adoptada para el festival es cell splitting hasta S4. Ambas decisiones se justifican porque son las primeras que satisfacen simultaneamente los criterios de cobertura, capacidad e interferencia bajo las hipotesis declaradas.

9.1 Nueva Pangea 2030

Si la densidad de usuarios creciera, la evolucion natural del proyecto apuntaria a small cells, mas espectro, beamforming, slicing y edge computing. El codigo del proyecto deja la base cuantitativa para ensayar esas extensiones.